

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)		(metals in Group 1) get more reactive (down the group) (1) award (1) for any of following due to a decrease in attraction between the nucleus and the outer shell electron easier to remove outer electron because there are more shells easier to remove outer electron because it is further from the nucleus	2			2		1
		(ii)		Group 1 metals are more reactive than Group 2 metals (1) award (1) for either of following - because Group 1 metals only need to lose 1 electron (from the outer shell) whereas Group 2 metals need to lose 2 electrons - because it is easier to lose 1 electron than 2 electrons	2			2		
	(b)			257.6 / 258 (3) if answer incorrect credit each correct step in one of two possible methods (ecf possible throughout) method 1 $n(\text{H}_2) = \frac{11.2}{2} = 5.6 \quad (1)$ $n(\text{Na}) = 5.6 \times 2 = 11.2 \quad (1)$ $\text{mass Na} = 11.2 \times 23 = 257.6 \quad (1)$ method 2 1 mol H ₂ produced by 2 mol Na / 2 g H ₂ produced by 46 g Na (1) 1 g H ₂ produced by 23 g Na (1) 11.2 g H ₂ produced by 23 × 11.2 = 257.6 g Na (1)		3		3	3	